

CSC 252 — Test 3 Study Guide (spaced / light)

Start Here

Guide updated: 2026-09-24 afternoon (America/Phoenix) -- Start Here rewrite. **LIGHT / spaced.** Reviewed 2026-09-24 evening by Claude Opus 5.5 (see §8c).

If you only read one thing: do almost nothing for Test 3 until Oct 8. The real risk is **Tue Oct 13**, a class day that stacks CSC 345 PP2 (2:00 PM), 252 HW3 (7:00 PM, rule-derived), and CSC 380 HW4 (11:59 PM, when released). Finish HW3 by **Mon Oct 12 night** so the Oct 13-14 review stays calm.

Next scheduled assessment

Field	Value	Label
Assessment	CSC 252 Test 3	confirmed item
Date	Thu Oct 15, 2026	confirmed (homepage)
When	During lecture period 9:30-10:45 AM (likely)	finer clock NV
Length	~25 min during lecture	likely -- Test 1 habit; not a published Test 3 cover
Coverage	not published	PROVISIONAL / TBD
Nearby syllabus decks	Deck 05 ALU (week of Sep 28), Deck 06 functions (week of Oct 5), Deck 07 formats (week of Oct 12)	schedule-based preview only -- not claimed as scope

Do **not** schedule study during class blocks (Tue/Thu through Dec 9, Arizona time): CSC 252 9:30-10:45 AM; CSC 335 12:30-1:45 PM; CSC 345 2:00-3:15 PM; CSC 337 3:30-4:45 PM; CSC 380 5:00-6:15 PM. All times below are **Arizona (MST, UTC-7)**. Combined daily order: `weekly/2026-09-24-weekly-study-order.md`.

Until Test 2 (Oct 1) is done, this guide stays light. Nearby: Sim 3 Wed Oct 7 7:00 PM (ALU project -- no AI help). HW3 is **rule-derived** Tue Oct 13 7:00 PM (not individually posted). Asm 3 ~Wed Oct 14 (syllabus "on or about").

Topics to study first (order + WHY)

1. **Official Test 3 topics (252-T3-tbd)** -- WHY: nothing is published; first job after each lecture from Oct 1 on is to write 2-3 bullets labeled **unverified**.
2. **Keep-warm Test 2 skills (252-T3-cont-binary, cont-mips, cont-gates)** -- WHY: same skills feed Sim 3 and later tests; **not** claimed as Test 3 scope.
3. **Optional preview Decks 05-07 (252-T3-preview / sec 4c)** -- WHY: those are the syllabus titles sitting in front of Oct 15. Use **only after the matching lecture**. No Sim 3 code.

Confirmed vs schedule-based: Date confirmed. Coverage is **not confirmed**. Anything from Decks 05-07 is schedule-based review, gated on lecture.

Day-by-day tasks (LIGHT; no class-block study)

When	Window	Task	Guide / slides / problems	Min
Thu Sep 24 - Wed Sep 30	After nearer Test 2 work only	Do not densify Test 3. Optional 10-min continuity flash.	P2 or GS-T3-L1	10
Thu Oct 1 after Test 2	10:45-12:30 gap (then other class blocks)	Write 2-3 unverified bullets from that day's lecture (likely ALU).	252-T3-tbd	15
Fri Oct 2 - Wed Oct 7	Spare 252 time belongs to Sim 3 (due Wed Oct 7 7:00 PM)	Almost nothing for Test 3. Optional 15-min continuity.	P3, P5	15
Thu Oct 8 - Sat Oct 10	After 335/345 exams	Add lectured notes to 252-T3-tbd; one MIPS or binary flash.	P4, P6	25
Sun Oct 11	Anytime	Mixed continuity + sec 4c for lectured decks only.	P2-P6; PV1-PV5 only if lectured	40
Mon Oct 12	Anytime	Structured review of known, lectured material only. Finish HW3 tonight (target) if it is posted.	All confirmed IDs	50
Tue Oct 13	Class day; HW3 7:00 PM if posted (rule-derived). Free gaps: before 9:30, 10:45-12:30, after 6:15	LIGHT only.	Flash cards	20
Wed Oct 14	No class; Asm 3 ~7:00 PM (approximate)	Calm recall. No cram.	One loop + one overflow	20
Thu Oct 15 morning	Before 9:30 AM	Buffer, then Test 3 in lecture.	--	10

Is this enough time? About 3.5 hours total. That is right for a light guide while scope is unpublished. **If scope is published and includes Decks 05-07** (ALU, functions, formats), make Sun Oct 11 and Mon Oct 12 deep days (about 90 minutes each). Use §4c PV1-PV5 plus that week's lecture examples.

Final self-check (spaced -- expand when scope is published)

- [] Homepage / tests/ folder re-checked for a published Test 3 scope
- [] Continuity set P1-P6 done blind
- [] Sec 4c used only for decks already lectured
- [] No Sim 3 code or hints taken from this guide
- [] Sleep the night of Oct 14

Slide excerpts

Decks 05-07 were **not** on the box harvest. After those lectures, review the live decks: - Deck 05 The ALU -- https://lecturer-russ.appspot.com/classes/cs252/fall26/slides/05_the_ALU.pdf - Deck 06 MIPS Functions -- [.../06_MIPS_functions.pdf](#) - Deck 07 Instruction Formats -- [.../07_instructionFormats_and_CPU_details.pdf](#)

For keep-warm binary/gates, reuse the Test 2 excerpts (Deck 01 p.41 invert-add-1; Deck 03 p.35 full adder) in [csc252/assessments/excerpts/](#).

Banner — scope PROVISIONAL / LIGHT. Test 3 scope is **not published**. The **date is confirmed**: Thu Oct 15, 2026, during lecture. Length is **likely** ~25 minutes (Test 1 pattern), not confirmed. The syllabus lectures just before Test 3 are Deck 05 (The ALU), Deck 06 (MIPS Functions), and Deck 07 (Instruction Formats and CPU Details). The course site does **not** assign those decks to any test, so this guide does **not** claim them as scope. Until coverage is published, this guide keeps Decks 01-04 skills warm and offers an **optional, clearly labelled preview** (§4c) to use only after lecture covers those topics.

Content-pass 2026-09-24: Content tree confirmed; PDFs already on box; no new invent beyond inventory.

Canonical maps: [csc252-topic-map.md](#) · [announced-assessments-inventory-2026-09-24.md](#) · [site-audit-2026-09-24.md](#)

1. Header

Field	Value	Status
Course	CSC 252 Computer Organization (FA26), TuTh 9:30-10:45 AM	Confirmed
Assessment	Test 3	Confirmed
Date	Thu Oct 15, 2026	Confirmed (homepage)
Length / when	~25 minutes during lecture; exact clock time not published	Length likely (Test 1 pattern); clock time NV
Coverage	not published	PROVISIONAL
Guide updated	2026-09-24 (America/Phoenix) — Opus review pass	

All times are **Arizona time (MST)**.

Nearby deadlines (energy planning only)

Item	When (Arizona)	Status
Sim 3 ("An ALU")	Wed Oct 7, 7:00 PM	Homepage + Gradescope
HW3	Tue Oct 13, 7:00 PM	Rule-derived from "due two days before each test". Not individually posted yet.

Item	When (Arizona)	Status
Asm 3	about Wed Oct 14, 7:00 PM	Syllabus says "on or about" — approximate

2. Topic checklist (light / TBD-aware)

Order	Section ID	Topic	Status
1	252-T3-tbd	Official Test 3 topics	TBD. Watch the homepage, tests/ folder, and lecture announcements.
2	252-T3-cont-binary	Continuation: binary, 2's complement, add/subtract	Keep warm. Not claimed as Test 3 scope.
3	252-T3-cont-mips	Continuation: MIPS loops and arrays	Keep warm. Not claimed as scope.
4	252-T3-cont-gates	Continuation: gates and adders	Keep warm. Not claimed as scope.
5	252-T3-preview	Optional preview of Decks 05-07 (§4c)	Use only after lecture covers it. Not claimed as scope.

When scope is published, replace row 1 and regenerate this guide.

3. Explanations (light)

Keep this short. Until coverage is published:

- Do a 15-minute flash on Test 2 skills each week, so they stay warm for Sim 3 and later tests.
- After each lecture from Oct 1 on, write down 2-3 topic bullets and label each **"unverified until slides/homepage confirm."**
- Use the official deck titles, not guesses about what might be tested.

How the preview works. The deck titles near Test 3 are public. §4c turns them into self-check prompts based on **standard MIPS conventions** (Patterson & Hennessy). Match notation to the actual slides, and skip anything not yet lectured.

Sim 3 boundary. Sim 3 ("An ALU") prohibits AI help. §4c covers ALU **concepts only**. It contains no Sim 3 code, structure, or hints.

4. Practice — continuity set (self-contained)

P1 (meta). What is confirmed about Test 3 today, and what is not?

P2 (binary add). Add 8-bit 0100 1011 + 0110 0110. Give the result, whether there is unsigned overflow, and whether there is signed overflow.

P3 (2's complement). Negate `0001 1100` in 8-bit 2's complement. Then sign-extend the result to 16 bits, in hex.

P4 (subtraction). Compute `0x50 - 0x68` in 8 bits using "invert, add 1, add". Give the signed result and say whether a borrow happened.

P5 (MIPS loop). Write a loop that sums the word array at `$s0` with length `$s1` into `$s2`, using only allowable-style instructions.

P6 (gates). Write Cout of a full adder as a sum of products, and say how many full adders a 32-bit ripple-carry adder needs.

P7 (planning). After Oct 1, list three lecture topics that might feed Test 3. Label each **unverified**.

4b. Light Gradescope-format continuity — Practice recommendation

This is inferred from the Test 1 result structure only: a scanned, hand-written paper of about 6 pages, with short-concept, binary-arithmetic, and C-to-MIPS questions. It is **not** a published Test 3 format.

GS-T3-L1. Round-trip unsigned 8-bit `0xD4` to binary, then to decimal, then back to hex. Show work as on a test page.

GS-T3-L2. In one line each: is `0xD4` negative as 8-bit 2's complement? What is its signed value?

GS-T3-L3. Write 3–5 MIPS lines that clear `$t0`, load word 2 of the array at `$s0` into `$t1`, and add it to `$t0`.

4c. OPTIONAL preview — Decks 05-07 (not claimed scope)

Only use these after the matching lecture has happened. Answers use standard MIPS and Patterson & Hennessy conventions. If the slides use different notation, follow the slides.

PV1 (Deck 05, ALU concept). A 1-bit ALU slice computes AND, OR, and add, then picks one result. Which circuit element selects the result, and what drives its select lines?

PV2 (Deck 05, ALU concept). How can an adder-based ALU compute $A - B$ without a separate subtractor circuit?

PV3 (Deck 06, functions). Which instruction calls a function, and where does it save the return address? Which instruction returns?

PV4 (Deck 06, functions). Which registers must a function preserve if it modifies them (callee-saved), and which may it overwrite freely (caller-saved)? Why does a function that calls another function need to save `$ra`?

PV5 (Deck 07, formats). Give the field layout and bit widths of the R-format, I-format, and J-format instructions.

5. ANSWER KEY

A1. Confirmed: Thu Oct 15, 2026, during lecture. Likely but not confirmed: ~25 minutes. Not confirmed: scope and the exact clock time.

A2. $0100\ 1011$ (75) + $0110\ 0110$ (102) = $1011\ 0001$ (177). - No carry out, so **no unsigned overflow**. - Both inputs are positive but the result's MSB is 1, so there is **signed overflow**. The true sum, 177, exceeds 127.

A3. $0001\ 1100 = 28$. Inverted: $1110\ 0011$. Plus 1: $1110\ 0100 = -28$ (0xE4). Sign-extended to 16 bits: 0xFFE4.

A4. $-0x68$: $0110\ 1000$ inverted is $1001\ 0111$, plus 1 is $1001\ 1000$ (0x98). Then $0x50 + 0x98 = 0xE8 = 1110\ 1000$. As a signed value that is $232 - 256 = -24$ (check: $80 - 104 = -24$). The carry-out is 0, so **a borrow happened**.

A5.

```
add $s2, $zero, $zero
add $t0, $zero, $zero
LOOP: beq $t0, $s1, DONE
      sll $t1, $t0, 2
      add $t1, $s0, $t1
      lw  $t2, 0($t1)
      add $s2, $s2, $t2
      addi $t0, $t0, 1
      j   LOOP
DONE:
```

A6. $C_{out} = A \cdot B + A \cdot C_{in} + B \cdot C_{in}$. A 32-bit ripple-carry adder needs 32 full adders. Bit 0 can be a half adder if there is no carry-in, but using a full adder there enables subtraction by setting $C_{in} = 1$.

A7. Your own list, with every item labelled unverified.

GS-T3-L1. $0xD4 = 1101\ 0100 = 128 + 64 + 16 + 4 = 212$, which is back to $0xD4$.

GS-T3-L2. Yes, it is negative (MSB 1). $212 - 256 = -44$.

GS-T3-L3.

```
add $t0, $zero, $zero    # t0 = 0
lw  $t1, 8($s0)          # word index 2 -> byte offset 8
add $t0, $t0, $t1
```

Preview keys (confirm against slides)

PV1. A multiplexer (MUX) selects the result. The operation / control bits drive its select lines.

PV2. Invert B (XOR each bit with 1, or use a "B-invert" MUX) and set the carry-in to 1. That computes $A + (\sim B) + 1 = A - B$, which is 2's complement again.

PV3. `jal label` jumps to the function and saves the return address ($PC + 4$) in `$ra`. `jr $ra` returns.

PV4. Callee-saved: `$s0–$s7` (plus `$sp`, and `$ra` in practice). Caller-saved: `$t0–$t9`, `$a0–$a3`, `$v0–$v1`. A function that makes a call runs `jal`, which overwrites `$ra`, so it must push `$ra` on the stack first and restore it before `jr $ra`.

PV5. Each format totals 32 bits.

Format	Fields (bit widths)
R-format	op (6), rs (5), rt (5), rd (5), shamt (5), funct (6)
I-format	op (6), rs (5), rt (5), immediate (16)
J-format	op (6), address (26)

6. Common mistakes + readiness

- Treating Test 2 topics, or the §4c preview, as confirmed Test 3 scope.
- Studying Deck 05–07 material before it has been lectured, or instead of Sim 3.
- Mixing up unsigned overflow (carry-out) with signed overflow (sign flip).
- Skipping sleep the night before Oct 15.

Readiness (spaced): 2–3 short sessions per week after Oct 8. Expand once scope is published or decks have been lectured.

Start Here (top of this guide) is the canonical day-by-day. The later prep table is kept as a short copy.

7. Day-by-day (spaced; no day-before cram)

Weekdays checked against the 2026 calendar.

Day	Plan
Fri Oct 2 – Wed Oct 7	Almost nothing for Test 3. Finish Sim 3 (due Wed Oct 7, 7:00 PM). Optional 15-minute continuity flash.
Thu Oct 8 – Sat Oct 10	Light: add lecture notes to 252-T3-tbd; one short MIPS or binary flash.
Sun Oct 11	Short mixed continuity set, plus §4c for any lectured decks.
Mon Oct 12	Structured review of known, lectured material only. Deep-ish day. Finish HW3 tonight if it is posted.
Tue Oct 13	LIGHT only — class day; HW3 due 7:00 PM (rule-derived).

Day	Plan
Wed Oct 14	Calm recall — no cram . Asm 3 due about 7:00 PM (approximate).
Thu Oct 15 morning	Buffer, then Test 3 in lecture.

8. Changes in this review (2026-09-24)

- Fixed the day plan's weekday errors. The old plan said "Mon Oct 13" and "Tue Oct 14" and had a stray "Wed Oct 14? wait" row; in 2026, Oct 12 is Monday, Oct 13 Tuesday, and Oct 14 Wednesday.
- Made the answer keys self-contained. The old keys said "see the Test 2 guide"; A2–A4 and L1–L3 now have full worked answers.
- Added the 25-minute length, clock-time-NV status, and nearby deadlines. HW3 is labelled **rule-derived**, and Asm 3 is labelled **approximate**.
- Added the optional §4c preview for Decks 05–07. It uses standard MIPS conventions, is explicitly **not** claimed as scope, is gated on lecture coverage, and gives no Sim 3 hints.
- Changed times to Arizona time.

8c. Opus 5.5 review (2026-09-24 evening, America/Phoenix)

- Re-checked every key (P2–P6, GS–T3–L1..L3, PV1–PV5) by hand and by script. **No wrong answers found.**
- Fixed a weekday error in Start Here: "Thu Sep 24 - Wed Oct 1" is now "Wed Sep 30" (Oct 1 is a Thursday).
- Length label changed from "confirmed" to **likely**.
- Flagged the Tue Oct 13 deadline stack and moved HW3 to a Mon Oct 12 night target.
- Added a scope-published contingency: two ~90-minute deep days if Decks 05–07 are named.
- Did **not** add any Test 3 scope. It is still unpublished.

9. Disclaimer

Study aid only — not for submitted work. Follow CSC 252 and UA academic-integrity and AI policies. Coverage is **provisional/TBD** until the instructor publishes it.